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SCREEN-LCA: Ex-ante prioritization of bio-based pathways for chemicals and plastics under a climate-mitigation objective

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ABSTRACT

Life cycle assessment (LCA) is a key tool for guiding circular bioeconomy strategies, including the substitution of fossil carbon with bioresources. However, its data- and time-intensive nature can hinder its use when there are numerous candidate pathways, particularly in the chemical sector. Here, we developed SCREEN-LCA, a stepwise ex-ante framework designed to prioritize case studies before conducting full LCA. The framework combines: (i) sector-level impact estimation; (ii) identification of hotspot products within targeted sectors; (iii) mapping of corresponding alternative bio-based pathways; and (iv) pathway prioritization based on multiple criteria. We applied SCREEN-LCA in a France-oriented demonstration to prioritize residual-biomass-based pathways for subsequent LCA. The greenhouse gas footprint of incumbent petrochemicals and plastics, feedstock compatibility, and technological maturity were used as screening criteria. An extensive graph database covering 50 hotspot products and their alternative bio-based routes was assembled. Ethylene, propylene, methanol, and PET emerged as priorities with promising bio-based alternatives.

1. Introduction

Climate change is primarily driven by the cumulative combustion of fossil resources, which remains the largest source of anthropogenic greenhouse gas (GHG) emissions (Calvin et al., 2023). In this context, bioeconomy solutions where fossil carbon is substituted with biogenic carbon have been put forward as a credible option to defossilize industrial systems and reduce associated GHG emissions (European Commission, 2025, 2012; Nature, 2026). To date, past policy efforts were mainly focused on increasing the use of renewable carbon in the energy sector (European Parliament, 2023a, 2023b). The petrochemical and plastic sectors have received comparatively less attention, despite growing recent interest (Buznitsky et al., 2024; European Commission, 2025). As of 2024, around 16 % of global fossil oil demand was used as feedstock for the industrial production of chemicals and materials such as plastics (IEA, 2025), and this is expected to rise. Fossil-based plastics alone are projected to account for roughly one-sixth of global oil demand by 2030 (IEA, 2025) and around one-fifth by 2050 (Suh et al., 2024). In fact, global plastic consumption is projected to triple between 2019 and 2060 (OECD, 2022). Reducing dependence on fossil resources therefore requires expanding the use of alternative carbon feedstocks for the production of polymers and organic chemicals. Bio-based alternatives also hold the advantage of temporarily sequestering biogenic carbon in products, which translates into climate benefits in terms of lowering global mean temperature change (Shen et al., 2023). In particular, the use of residual biomass, which constitute often underused streams not primarily produced for food, feed, material or energy purposes, may improve resource efficiency and reduce pressure on land compared with purpose-grown biomass (European Commission, 2025).

From a sustainability perspective, efforts to develop a bio-based chemical sector should focus on providing substitutes to products associated with the highest environmental burdens, as promoted by the EU's Safe and Sustainable by Design framework (JRC, 2024). The identified bio-based alternatives have to be feasible, technically and resource-wise (European Commission, 2025). However, petrochemicals and plastics comprise a wide range of products, including thousands of molecules (Speight, 2019) and dozens of polymer families (ISO, 2011;

Abbreviations: GWP, Global warming potential, generally evaluated as GWP100 with a 100-year time range; GHG, Greenhouse gas; LCA, Life cycle assessment; LCI, Life cycle inventory; PET, Polyethylene terephthalate; PLA, Polylactic acid; EU27, European Union and its 27 member states as of 2020; SM, Supplementary materials; BTX, Benzene, Toluene and Xylenes; EE-MRIO, Environmentally extended multi-regional input-output; TRL, Technology readiness level.

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ASTM, 2018; OECD, 2025a). Choosing which alternatives to develop is therefore far from trivial. Prioritization is especially crucial when leading environmental assessment methodologies, such as life cycle assessment (LCA) (ISO 14040:2006), remain data- and time-intensive (Beemsterboer et al., 2020; Saavedra-Rubio et al., 2022).

A handful of attempts has been proposed to prioritize bio-based alternative products, including approaches designed to pre-select promising candidates for subsequent LCA (de Ruiter et al., 2025; European Commission, 2019; Winter et al., 2022). However, these studies either cover a narrow portion of the chemical and polymer landscape, or rely on selection criteria that are insufficiently transparent to enable reproducibility. For instance, the early U.S. Department of Energy screening of 'top-value added chemicals' (Werpy and Petersen, 2004) or the more recent work of Winter et al. (2022) focus on sugar-based conversion routes, leaving non-carbohydrate feedstocks aside. In contrast, other methods offer broader coverage (de Jong et al., 2012; European Commission, 2019; IEA Bioenergy, 2020), but their selection procedures are only partially documented, which makes replication difficult, if not impossible. On the other hand, the work of Suh et al. (2024) and de Ruiter et al. (2025) present methods that are intended to be reproducible and applicable to a broader panorama of feedstocks. However, they are respectively specific to finding plastic materials causing high environmental burdens or alternatives to fossil-based plastics, excluding other petrochemicals. Based on this state-of-the-art, no existing reproducible method appears to be available with broad coverage, encompassing various feedstocks and products, for prioritizing efforts to identify the most promising bio-based alternatives to petrochemicals and plastics.

To bridge this gap, this study introduces SCREEN-LCA, a stepwise, reproducible and sector-agnostic methodological framework for ex-ante prioritization of alternative bio-based pathways. The framework is designed to support decision-making by identifying priority candidates whose environmental performance remains to be quantified through detailed LCA. First, high-impact incumbent products are identified based on a selected impact indicator, such as their GHG footprint. Second, candidate alternatives are screened according to feedstock compatibility and technological maturity. Where multiple routes exist, the framework prioritizes the more technologically mature options under the selected decision context. We demonstrate SCREEN-LCA by applying it to the French petrochemical and plastic sectors in order to identify promising bio-based pathways capable of valorizing residual biomass with strong climate change mitigation potential.

2. Method and demonstration

The prioritization framework consists of eight steps depicted in Fig. 1. It can be applied to various impact-intensive sectors (or products).

2.1. Step 1: contextualization and scoping

This initial step sets the context and the scope of the study, thereby involving normative choices. It draws inspiration from the goal and scope definition phase of the LCA methodology standards comprising the ISO 14040 and 14044 (ISO, 2006a, 2006b). An example of goal could be as follows: "*identify and prioritize alternative construction materials with improved energy performance in Europe*". In relation to the defined goal, a series of scoping conditions are established to delineate the boundaries of the chosen case study, as highlighted in Table 1.

For the demonstration of SCREEN-LCA presented herein, the goal is to "*prioritize bio-based pathways using residual biomass capable of substituting petrochemicals products and plastics with high GHG footprint in France*". Accordingly, candidate pathways should be compatible with the type and quantity of residual biomass available in France and have the potential for large-scale deployment by 2050.

Here, we sought to identify high-GHG-footprint products among both domestically produced and imported goods, thereby capturing product flows that are potentially addressable through national action. Inclusion of imported goods broadens the scope beyond domestic production, acknowledging that national policies can influence product flows not only through production-side measures but also through procurement, trade, and demand-side interventions. With this screening boundary, the quantity considered is therefore defined as domestic production plus imports; exports are not subtracted. Therefore, this quantity should not be interpreted as apparent domestic consumption. Due to data availability, the application domain is restricted to single-compound (i.e., pure substances), thereby excluding formulations and composites.

2.2. Step 2: estimation of the baseline footprint of the target sectors

This step's objective is to estimate the baseline impact of the target sectors (scoping condition from step 1) and to compare it to a reference footprint (e.g., region's consumption footprint or direct GHG emissions). Thereby, it permits to justify the relevance of the selected goal and scope or to adjust it before putting too much effort in the wrong direction.

Useful data to estimate sectorial impacts lie in environmentally extended multi-regional input-output (EE-MRIO) databases. For the

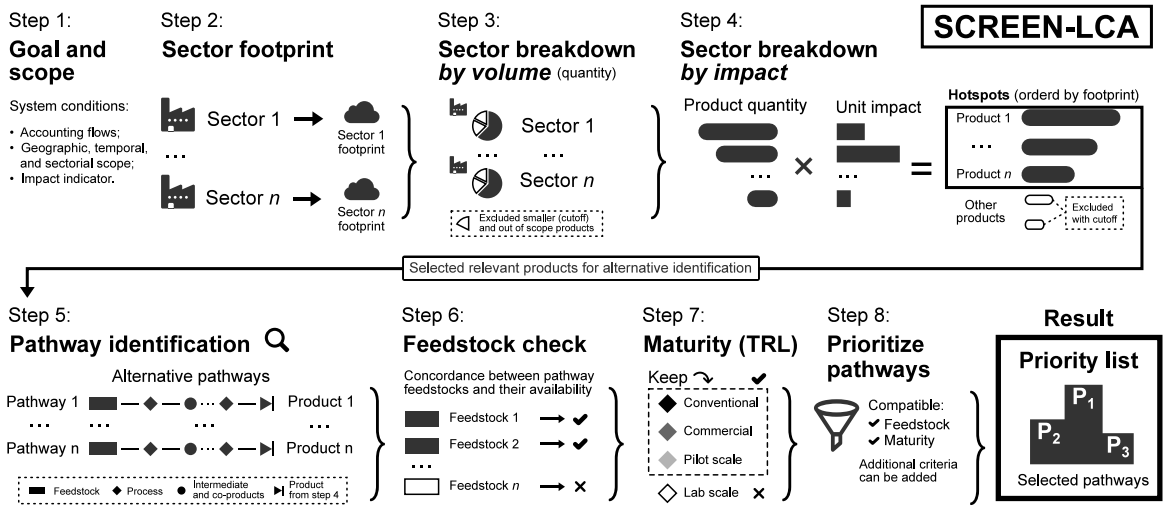


Fig. 1. General overview of SCREEN-LCA, the stepwise ex-ante prioritization framework of alternatives to high-environmental-impact products.

Table 1
Scoping conditions to be specified in step 1, once the goal is defined. A non-exhaustive list of examples is provided, together with the specific conditions applied for the illustrative demonstration (see 'this study'). For most scoping conditions, combinations are possible (e.g., several geographic regions, products, indicators).

Scoping condition	Description	Examples
Application domain	Defines the possible sectors targeted for bio-based alternative applications	• Pure petrochemicals and plastics (this study); • Energy; • Food; • Transportation.
Temporal scope	Baseline year and period of the study over which the goal applies	• Baseline: short-term (e.g., 2019–2024 average to represent 'present') (this study); • Prospective study: mid-term (e.g., year 2040); • Prospective study: long-term (e.g., year 2060).
Geographical scope	Geographic area of interest is defined[1]	• France[2] (this study); • Lombardy Region, Italy; • European Union, as a whole, or for all/specific (e.g., NUTS-3 regions of the EU) (European Parliament, 2003); • World.
Impact indicator	Metric aligned and coherent with the goal used for the screening of hotspots	• GHG footprint, GWP_{100} (demo); • Economic indicators (e.g., value-added, GDP, growth); • Social indicators (e.g., jobs, salary, working conditions); • Other environmental indicators (e.g., typical LCA category indicators, endpoints, aggregated score[3]).
Alternative feedstock	Type of feedstock considered for the alternatives	• Residual biomass (this study); • Captured carbon (e.g., DACCU[4], PSCCU[5]); • Other waste recycling (e.g., municipal solid waste, construction waste).
Accounting system, baseline	Accounting approach used to estimate the impact of the fossil-based reference footprint	• Footprint of produced and imported petrochemicals and plastics (this study); • Direct emissions per sector; • Imports and exports footprint.

[1] Depending on data accessibility, a geographic scope using typical statistical boundaries (e.g., country, economic group of countries) is recommended.
[2] Metropolitan France excluding overseas departments and territories, including Corsica.
[3] See ISO 14044:2006 (ISO, 2006b).
[4] Direct air carbon capture and utilization (DACCU).
[5] Point-source carbon capture and utilization (PSCCU), e.g., bioenergy or cement clinker production.

demonstration, we mobilized EXIOBASE (Stadler et al., 2018). It provides global coverage and consistent data across countries, sectors and products and tracks the major GHG. We took the most recent version of EXIOBASE (v3.10.2), more specifically the 2022 data using input-output tables (IOT) in the product-by-product format (i.e., "*IOT_2022_pxp*"). This was the most recent time series to avoid using now-casted values, despite data going up to 2024 (Stadler et al., 2026). "*Plastics, basic*" and "*Chemicals nec*" are the two EXIOBASE products matching with the study's scope (step 1). For the demonstration, the GHG footprints of "*Plastics, basic*" and "*Chemicals nec*" were estimated for the same screening boundary defined in step 1, i.e., domestic production plus imports, using the GWP100 indicator (Table 1). Note that EXIOBASE provides multiple socio-economic and environmental data, making it possible to derive other indicators (e.g., water and energy uses, air emissions, employment). EXIOBASE calculation details specific to the demonstration are available in the SM's section 2.2.

As a result, the GHG footprints of the produced and imported plastics ("*Plastics, basic*") and other chemicals ("*Chemicals nec*") in France in 2022 are estimated at 21.7 and 22.2 Mt of CO_2-eq respectively, underscoring the importance of target sectors. In comparison, the total direct GHG emissions and consumption footprint of France can be estimated to 390 and 592 Mt of CO_2-eq, respectively, using the same EXIOBASE version and for the same period (see SM 2.3). These results are in agreement with official statistics (Citepa, 2025). The estimated footprints also demonstrate that France has direct control over some high-GHG-footprint industrial products, thereby justifying the pursuit of bio-based alternatives. This trend is generally true across the European Union, as shown through sensitivity analysis (see SM 5.1), or within specific studies (Wood et al., 2020).

2.3. Step 3: refining the product scope through a breakdown of target sectors

After confirming the relevance of the goal and scope with the baseline impact estimation, step 3 disaggregates the targeted sectors into specific and isolated products based on their physical market share. The breakdown in physical units also facilitates subsequent footprint estimations (step 4) as mass is generally the standard unit in LCA databases (e.g., Ecoinvent, Sphera). To this end, statistical databases documenting physical flows such as PRODCOM (Eurostat, 2025, 2023) and BACI (CEPII, 2025; Gaulier and Zignago, 2010) can be mobilized. They provide data at detailed product levels (Table 2). After data extraction and processing, products are ranked in decreasing order of mass. SCREEN-LCA demonstration uses Eurostat PRODCOM dataset "ds-059358" to decompose the targeted petrochemical and plastic sectors (Eurostat, 2026) into specific products. This database provides several useful indicators. It reports the production and imports values separately in physical unit, which suits the scope. However, data gaps are frequent for France so aggregated EU27 data is used as a proxy. Although France's petrochemical share in European production capacity is about 13.4 % (JRC, 2021), this proxy is seen as the best balance between data availability and accuracy of regional specificities. To a minor extent, the EU27 level also presents some gaps depending on the product and the year. The PRODCOM database covers a broader range of products than the scope of the demonstration and need filtering. The products belonging to the target sectors are identified based on the classification of product by activity (CPA) code, which corresponds to the first 6 digits of the PRODCOM code. Imported and production quantities for each product were added together, the resulting list was sorted by decreasing order of magnitude. We explain in more details how data was handled in SM's section 3. The reproducible Python code and resulting data are available in the Git repository.

2.4. Step 4: hotspot products identification

Building from the data obtained at step 3, hotspots products are here identified and efforts of the subsequent steps will focus on them. Each product is checked to remove those falling outside the study's scope. Then, the corresponding impact data are retrieved from a relevant life cycle inventory (LCI) database (e.g., Ecoinvent). For products with no matching LCI dataset, appropriate proxies in the LCI database or scores retrieved from the LCA literature can be used. If no suitable proxy can be found, the product may be excluded assuming relatively lower importance. The indicator used to estimate the impact should be consistent with the one defined at step 1 (e.g., GWP_{100} in CO_2-eq for climate change impact). Given the time-intensive nature of this process, proper screening and sorting are paramount in previous steps, especially in step 3.

Several products remained after the classification-based filtering in the demonstration's step 3, notably some leftover inorganic (e.g.,

Table 2
Non-exhaustive list of useful databases for use in step 3, with their key characteristics.

Database[1]	Geographical scope	Flows considered	Reporting unit(s)	Strengths	Drawbacks	Reference
Eurostat **PRODCOM**	EU + Europe	production (sold or actual), detailed as controlled and subcontracted, impots, exports	value (EUR €) and physical units (e.g., kg, m, m^2, m^3, number of items, …)	High consistency in accounting. Limited to EU studies. Many items are accounted with the same physical unit. Production and trade data. High granularity.	Some products (e.g., consumer products) are expressed in a wide variety of units	(Eurostat, 2025, 2023)
UN **Comtrade**	Global, >130 countries	Trade	value (USD $), sold units and physical units (e.g., kg, g, TJ, m^3, km, …)	Show inconsistencies (classification, confidential values, and some year gaps, import/exports differences)	Only trade	(UN, 2025)
CEPII **BACI**	Global, >200 countries	Trade	value (thousand USD $) and physical unit (i.e., tonnes)	Similar to Comtrade, but corrected for more consistency in data. High granularity.	Only trade	(CEPII, 2025; Gaulier and Zignago, 2010)
OECD **Trade by Enterprise Characteristics (TEC)**	Global	Trade	value (USD), number of enterprises		Only trade	(OECD, 2025b)
Any national and/or sector specific databases.	Country	-	-	Ideal when other databases are not fitting the need.	Datasets may not be organized and aggregated in the same way across regions.	-
US EPA **Chemical Data Reporting (CDR)**	USA	Production	physical production (i.e., pounds)	High granularity.	Specific to chemicals	(US EPA, 2022)
OECD **Plastics Use in 2019**	Global	Use	physical volumes (i.e., million tonnes)	Specifically covers plastics, with a consumption approach that is ideal as part of footprint. Limited granularity in terms of world regions and plastic types. High granularity.	Specific to plastics	(OECD, 2024)

[1] Institutional reference with specific database name in bold; CEPII: Centre d'Études Prospectives et d'Informations Internationales; UN: United Nations; OECD: Organisation for Economic Co-operation and Development, US EPA: United States Environmental Protection Agency.

activated natural mineral products) or already bio-based products (e.g., cellulose and its derivatives), and manual screening was required. More importantly, we excluded a series of products that are considered too broad. For example, "*Other oils and oil products, n.e.c.*" is the highest-ranking product by mass in the EU27, but it represents a heterogeneous mix rather than a specific product. Exclusions are further documented in the SM's section 3.3 and Step 3 Excel file.

There are various databases from which the corresponding impact factors can be derived. Ecoinvent database was judged relevant for the demonstration, due to its wide coverage of industrial products in the European context (Wernet et al., 2016). We used Ecoinvent version 3.12 with the cutoff allocation model (Ecoinvent, 2024) to estimate products footprints. We performed a cradle-to-grave analysis, from the raw material extraction to the final disposal, but we excluded the use phase impacts that are assumed unchanged in this drop-in substitution approach. There is currently no straightforward method readily available to match the products' volume data with the corresponding Ecoinvent datasets (i.e., unit process) and geographies. Consequently, this manual task relied on PRODCOM product names and the detailed documentation available in the Ecoinvent database. Ecoinvent provides some market representative datasets which represent a realistic combination of geographical origins and processes. These, labelled as "*market for …*", when available they were preferred over specific datasets. The market's supplying geographies were estimated using a combination of PRODCOM and BACI data (Table 2) and the procedure is described in SM section 4.2. When Ecoinvent was more specific than PRODCOM, the product was either disaggregated into several products, or a representative dataset was identified for the family. For example, the PRODCOM product "*Phthalic anhydride; terephthalic acid and its salts*" is matched with Ecoinvent's "*market for purified terephthalic acid*". After matching the top PRODCOM products with Ecoinvent datasets, their climate change impacts were estimated via the GWP_{100} indicator derived from the IPCC 2021 (AR6) method, ensuring consistency with step 2. The end-of-life footprint was estimated by approximating all products to plastics, considering that they were mostly intermediates and using OECD's data on plastic end of life (OECD, 2022) (SM 4.4). We judged appropriate to also estimate the upstream contribution of the following key intermediates in the overall hotspot product footprints: ethylene, propylene, benzene-toluene-xylenes (BTX). Indeed, they are prominent petrochemical intermediates (Petrochemicals Europe, 2019; IEA, 2018). The matching choices are available in the Step 3 Excel SM file, and the footprint calculation method is provided in a documented Python notebook (available in the Git repository). Results of the top 15 hotspot products are presented in Fig. 2, and more detailed data is in Step 4 Excel SM. Remember that this is a hybrid approach, using EU27 data as a proxy for France's petrochemical industry. In addition, sensitivity analysis was performed to test the results (see SM section 5.3).

2.5. Step 5. alternative pathways identification

Step 5 explores possible bio-based alternative pathways for each of the 'hotspot products' identified in step 4 (illustrated in Fig. 2), primarily through a structured literature review. For the illustrative demonstration, the results of this step are structured as a visual and modular database linking hotspot products (e.g., bio-based ethylene), intermediates (e.g., ethanol) and feedstocks (e.g., cereal straw). Intermediates are connected to at least one upstream and one downstream product via a conversion process (e.g., ethanol dehydration).[1] This representation captures the diversity of bio-based pathways for each targeted product and reveals common intermediates and processes.

[1] Available at https://kumu.io/nicolas-lienart/screen-lca-bio-based-pathways-database (also available as JSON and Excel in the Git repository).

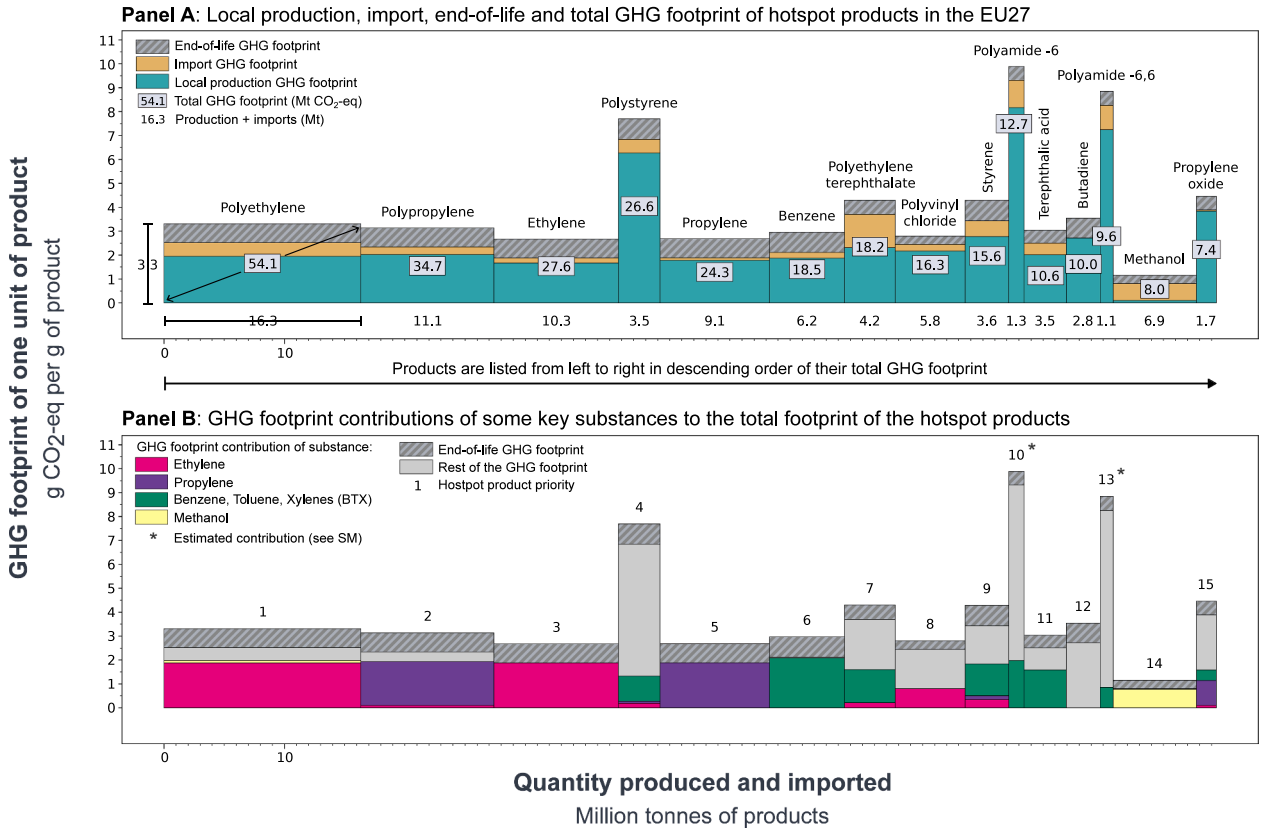


Fig. 2. Top 15 hotspot EU27 petrochemicals and plastics ranked by total GHG footprint.
Each bar width reflects volume from PRODCOM data, and height reflects unit GHG footprint (production and end-of-life, IPCC 2021 GWP_{100}) derived from Ecoinvent v3.12 (cut-off). Bar area represents total estimated footprint. **Panel A** specifically shows the EU27 production, imports, end-of-life footprints as areas and indicates the total GHG footprint and available volume. **Panel B** shows the footprint contribution of some selected substances and indicates the GHG ranking of hotspot products. See Step 4 Excel SM and Git repository for the background data and code.

Complete pathways from feedstock to the targeted products are formed determined by all the possible paths going from input to hotspot products.

The literature search was conducted using Google Scholar with queries in the following form: *(("chemical_IUPAC_name" [OR "chemical_alias_1" …]) AND ("bio-based" OR "biobased" OR "Biotechnological production" OR "from biomass" OR "bio-derived" OR "bio-based alternatives")*. Quotes ensure exact matches and optional aliases *[OR "chemical_alias_1" …]* can capture usual synonyms which can be found on PubChem (Kim et al., 2025). Additional references were collected to complete data gaps, including Clarivate Web of Science and Google Scholar email alerts, publications references and technical websites. Data was organized using Kumu,[1] which is an online relationship-based mapping tool allowing efficient querying, maintenance and visualization of the data from an end-user perspective. The resulting graph database represents the 50 hotspot products from step 4 with additional products (mostly intermediates) totalling to >100 products. In between, we have documented >100 intermediate processes, which interconnect products through directed edges. Each node, whether a product or a process, is annotated with some metadata, including processes' literature references (SM 7.2). The raw data is also available in the Git repository.

2.6. Step 6. feedstock compatibility

The pathway screening must align the selected feedstock of the goal and scope (step 1) to supply the pathways highlighted in step 5.

The demonstration is restricted to the residual biomass available in France. To assess the compatibility of pathways uncovered in step 5 with this scope, the inventory of theoretical residual biomass potential generated in France was used (Javourez et al., 2024). The total residual biomass potential (dry mass) is estimated at 107.1 Mt · y^{-1}, mainly composed of 67 % crop residues, followed by 13 % animal manure and 7.8 % for forestry residues. The demonstration focused on lignocellulosic materials, given that they largely constitute the most voluminous residual resources (i.e., crop residues).

2.7. Step 7. processes technological maturity

The technological maturity of each conversion process identified in step 5 is evaluated, based on literature data. The assessment is restricted to processes compatible with the feedstock selected (step 6). Process maturity can be ranked using the technology readiness level (TRL) framework (Mitchell, 2007). Where relevant, a cutoff (e.g., TRL above 4) can be applied as threshold to filter out processes. When TRL are not explicitly reported, they are inferred from the descriptions provided in the literature, in particular statements about development or commercialization status.

The demonstration revealed that some technologies, although frequently advertised, have not reached maturity. This is notably the case for routes relying on complex catalytic conversions where catalysts are still under development (Wang et al., 2023; Wu et al., 2023). In addition, biological production routes involving genetically engineered microorganisms are still not mature (nova-Institute, 2013; Rosenboom et al., 2022; Toor et al., 2020; Tse et al., 2021). Building from the TRL framework, four maturity levels were distinguished (SM Table S3):

- "Low": laboratory- or bench-scale processes;

- "Medium": pilot-scale, including industrial demonstrations;
- "High": commercial technologies with emerging markets;
- "Fully mature": widespread commercial processes with large production.

This information, together with the supporting references, is added to the graphical database of step 5.[1] Fig. 3 specifically illustrates step 6 and 7 for the case of bio-based propylene.

2.8. Step 8. building the priority list and results

The final priority list is established by integrating the outcomes from step 4 to 7. Hotspot products are considered sequentially, in decreasing order of footprint. When a product is directly derived from a single intermediate, this intermediate is prioritized instead. For example, in the demonstration, polypropylene and polyethylene were excluded in favour of propylene and ethylene, their respective monomer. The maturity score of the pathway is defined by the minimum process maturity found along the conversion chain. When two pathways share the same overall maturity, individual process maturity and the bottleneck positions are used to discriminate, on a case-by-case basis, the route that is the most mature.

For the illustrative demonstration, the key hotspots and their alternatives priority list is as follows:

i. Ethylene ranks first. The ethanol and the methanol bio-based routes both stand out to produce bio-based ethylene (Fig. 3). These alternatives are compatible with the lignocellulosic-rich feedstock selected (step 6), they reached medium-to-high maturity across the entire pathway. However, the ethanol pathway currently shows more commercial prospects (ADEME, 2024). Therefore, the ethanol-to-ethylene pathway is prioritized for further validation via LCA. Ethylene is also an intermediate, feeding several downstream hotspot chemicals (e.g., ethylene oxide, ethylene glycol) (IEA Bioenergy, 2020). It is a highly connected upstream within the hotspot database.
ii. Propylene ranks second. Bio-based propylene can be produced from methanol using the methanol-to-propylene (MTP) or the methanol-to-olefin (MTO) processes (Zimmermann, 2013). Like ethylene, it includes a wide portfolio of downstream hotspot chemicals (e.g., acrylic acid, propylene oxide) (IEA Bioenergy, 2020). The MTO/MTP pathways have reached industrial pilot scale (Tian et al., 2015). Bio-based methanol can be produced at pilot-scale (IEA Bioenergy, 2020) via biomass gasification, followed by syngas conversion (Olah et al., 2009). Methanol itself is listed among the top hotspots (Fig. 2)
iii. The third position is for PET, a major hotspot product. The polylactic acid (PLA) is identified as a relevant substitute. PLA is derived from fermentation to lactic acid and subsequent polymerization (IEA Bioenergy, 2020; Lee et al., 2022). PLA is the most established bio-based plastic in terms of industrial deployment, with demonstrated commercial-scale production. In contrast, polyethylene furanoate (PEF) remains far less mature despite strong potential (Rosenboom et al., 2022). However, PLA might be able to substitute PET in some situations, but not for all of PET's applications (Sousa et al., 2021).

Benzene, toluene and xylenes (BTX) aromatics and their derivatives were also identified as key hotspot chemicals (Fig. 2), entering the composition of polystyrene, styrene, PET, terephthalic acid and polyamides −6 and −6,6. They can be obtained from catalytic fast pyrolysis (Wegelin and Meier, 2024) and methanol to aromatics (IEA Bioenergy, 2020; IEA, 2018). However, the lower technological maturity of these routes (Wu et al., 2023) limits their prioritization under the criteria applied here.

Overall, the resulting priority list is consistent with geopolitical and industrial concerns in Europe. Notably, a recent joint statement published by seven European countries listed ethylene, propylene and methanol as strategic for the chemical industry, methanol was highlighted as particularly critical (Ministry of the Economy, Finance and Industrial and Digital Sovereignty, 2025).

3. Discussion

3.1. Method-related considerations

Despite the demonstration's focus, SCREEN-LCA was designed as a generic and configurable framework. It can accommodate different goal and scope, making it unrestricted to residual biomass uses in France or to the GHG emission indicator. In principle, any metric that can be quantified consistently at step 2 and 4 can be used in the ranking of hotspots, including socio-economic indicators. SCREEN-LCA can be extended beyond biomass to other feedstock classes (e.g., waste-derived resources and captured CO_2).

Screening a single driving environmental indicator at a time is a deliberate feature of SCREEN-LCA. Consequently, unlike LCA, it is not designed to uncover impact trade-offs. It should be used as a complementary tool to prioritize pathways before conducting full LCAs.

Some of the initial steps of the method are largely based on publicly available institutional data depicting the current situation. By contrast, the steps related to alternatives (step 5 to 7) mostly use scientific literature and institutional reports. To simplify and further accelerate the process, these data could be obtained from expert interviews and selective use of grey literature (e.g., industry briefing, trade press). However, this would likely increase subjectivity and reduce reproducibility. Step 7 is already a weak part of the method in this sense due to the need to interpret current literature regarding the technological maturity of processes and more systemic data would allow for more

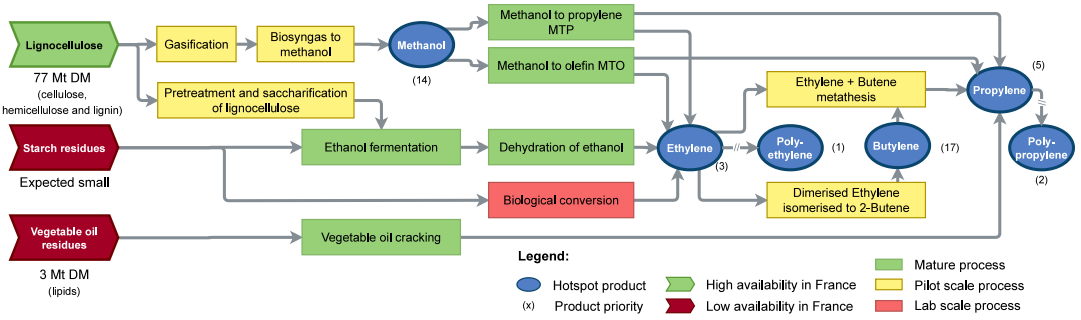


Fig. 3. Illustrative overview of the bio-based propylene pathways (step 6), considering the residual biomass feedstock availability in France (step 5) and the technological maturity (step 7). Only the relevant processes are depicted. Key intermediates (methanol, ethylene and butylene) are represented but co-products are not depicted for readability. Feedstock potentials are derived from (Javourez et al., 2024). Priority codes correspond to the hotspot product priority rank available in the Step 4 SM Excel, also visible in the Fig. 2B.

robust conclusions. Anyway, clear and transparent documentation of the procedure and assumptions should always be provided when the results are communicated.

Some limitations of SCREEN-LCA, besides data-dependency, delineate avenues for refinement. First, demand is treated as exogenous: the method does not represent consumption dynamics, substitution behaviour or policy-driven demand shifts. Future work could integrate alternative demand trajectories, for instance through scenario analysis reflecting behavioural change, regulation, or market transformation. Second, prioritizing the most mature routes may underweight early-stage breakthrough options that are currently rated with a low TRL while they could deliver superior long-term performance (Buznitsky et al., 2024). This trade-off could be addressed by running the screening under alternative maturity thresholds, or by complementing "deployment-ready" prioritization with a parallel "high-potential" innovation shortlist. Finally, the end-of-life footprint is accounted but simplified due to methodological challenges of estimating the fate of the different petrochemicals. Though the estimated relative effect of the end of life on the hotspot priority is limited. Studies with more contrasted products should adopt more precise estimations when judged relevant.

3.2. *Insights from the demonstration*

A notable outcome of the demonstration is that several widely discussed bio-based plastics and platform chemicals did not emerge among the highest priorities, although for different reasons. Some typical example like succinic acid, glycerol, furfural, levulinic acid, hydroxypropionic acid, sorbitol or xylitol (Bozell and Petersen, 2010; IEA Bioenergy, 2020; Isikgor and Becer, 2015; Werpy and Petersen, 2004) were not associated with the highest-ranking incumbent hotspots. Whereas alternative bio-based plastics such as PEF and PHA (Isikgor and Becer, 2015; Rosenboom et al., 2022; Werpy and Petersen, 2004) were identified as possible alternatives but ranked below more mature pathways. This reflects the starting point of SCREEN-LCA: rather than beginning with a predefined portfolio of bio-based products, the framework first identifies incumbent petrochemicals and plastics associated with the largest GHG burdens to date and proceeds to find alternative pathways capable of substituting these hotspots. Candidate pathways are subsequently screened for compatibility with the selected feedstock scope and for technological maturity. Prominence in the bioeconomy literature therefore does not, by itself, determine priority for subsequent LCA. Instead, SCREEN-LCA implements a problem-oriented sequence in which the environmental burden to be addressed precedes the selection and screening of candidate solutions. The sensitivity analysis (SM section 5) further showed that the identification of the leading incumbent hotspots was relatively stable across the tested data configurations.

3.3. *Future perspectives*

While some efforts were required to produce the database of bio-based pathways behind this article, future research may reuse, build upon and consolidate it. This can be achieved not only by expanding the scope but also by updating the data. This database of pathways can evolve into a robust overview tool for the bioeconomy, facilitating screening and research prioritization.

The prioritized list of chemicals and intermediates identified through the proposed framework provides a clear roadmap for conducting state-of-the-art LCA studies. These assessments should estimate precisely the environmental performance of the most promising alternative pathways compared to the fossil-based reference products.

Finally, all the code used for producing the results is made public, allowing reuse, reproduction and expansion of this work.

4. Conclusion

Robust life cycle assessments are inherently time- and data-intensive. This makes transparent, defensible case study selection a prerequisite whenever the number of possible products and alternative routes is large, like illustrated with the chemical sector. Yet, structured and reproducible approaches to prioritize mitigation options and compare technological pathways remain limited. This study addressed this gap by proposing and implementing SCREEN-LCA, a stepwise and operational LCA case prioritization framework based on explicit and justified assumptions.

The method was illustrated through a France-oriented case study of residual-biomass-based alternatives to petrochemicals and plastics. It ranked incumbent products according to their baseline GHG footprint and screened candidate bio-based pathways according to feedstock compatibility and technological maturity. Ethylene and propylene emerged as priority hotspot products with comparatively mature candidate routes, including (i) ethylene from second-generation ethanol fermentation and (ii) propylene via biomass gasification, methanol synthesis, and methanol-to-olefins. PLA was also identified as a candidate alternative for selected PET applications. These pathways are priority candidates for subsequent full LCA, which is required to quantify their environmental performance and climate-mitigation potential.

Beyond the prioritized list of pathways, a major contribution of this work is the development of a graph database of bio-based value chains. The database interconnects feedstocks, intermediates, processes and end products, with documented maturity ratings and supporting bibliographic references. By structuring more than 50 hotspot products with their alternative production routes, the graph database provides a reusable knowledge base for systematic screening of bioeconomy solutions.

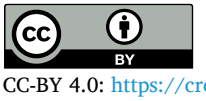
The SCREEN-LCA method is not restricted to the prioritization of bio-based alternatives to fossil-based chemicals and materials under a GHG mitigation objective. It can also be adapted to other geographical contexts, time horizons and strategic goals (e.g., circularity, resource efficiency, or alternative feedstocks). As such, this research offers the methodological foundation for reproducible ex-ante case studies selection for full LCA. In addition, the illustrative demonstration provides specific data to inform strategic decision-making in the transition towards bio-based products.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used OpenAI's ChatGPT, Google's Gemini and Anthropic's Claude to provide suggestions for adjusting the abstract length according to the journal's word limit and to assist in the coding. Every use of the LLMs is reported in the code located in the Git repository, along with a public link to the chat. The use of generative AI remained strictly limited. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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CRediT authorship contribution statement

Nicolas Lienart: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Thibaut Lecompte:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Lorie Hamelin:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.resconrec.2026.109198.

Data availability

A PDF document is provided as supplementary materials on the article's online page, along with three Excel files, referred in the article as Step 2, Step 3 and Step 4. All the code produced and used for this study is available in the INRAE Git repository: https://forge.inrae.fr/nicolas.lienart/screen-lca-paper-supplementary-code/. The same material is also hosted on GitHub: https://github.com/nicolnt/screen-lca-paper-supplementary-code.

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